Calculus I

Computation problems #7

Topics:

- Limits of finite sums
- Definite integrals
- The Fundamental Theorem of Calculus
- Substitution method
- Areas Between Curves

Problem 1. Graph the function and find its average value over the given interval.

$$f(x) = -3x^2 - 1, \quad [0, 1].$$

For Problems 2–4, suppose that f and g are integrable and

$$\int_1^2 f(x)\,dx = -4, \quad \int_1^5 f(x)\,dx = 6, \quad \int_1^5 g(x)\,dx = 8.$$

Find the given integrals.

Problem 2.

(a) $\displaystyle\int_2^2 g(x)\,dx.$

(b) $\displaystyle\int_5^1 g(x)\,dx$

Problem 3.

(a) $\displaystyle\int_1^2 3f(x)\,dx$

(b) $\displaystyle\int_2^5 f(x)\,dx$

Problem 4.

(a) $\displaystyle\int_1^5 [f(x) - g(x)]\, dx$

(b) $\displaystyle\int_1^4 [4f(x) - g(x)]\, dx$

For Problems 5–6, use the FTC to find dy/dx, if

Problem 5.

(a) $\displaystyle y = \int_a^x (t^3 + 1)\, dt$

(b) $\displaystyle y = \int_x^5 3t \sin(t)\, dt$

Problem 6.

(a) $\displaystyle y = \int_1^{x^2} \cos(t)\, dt$

(b) $\displaystyle y = \int_{1+3x^2}^4 \frac{1}{2+t}\, dt$

For Problems 7–10, find the given integrals.

Problem 7.

(a) $\displaystyle\int (x^3 + x)^5 (3x^2 + 1)\, dx$

(b) $\displaystyle\int \sqrt{2x+1}\, dx$

Problem 8.

(a) $\displaystyle\int \sec^2(5x + 1) \cdot 5\, dx$

(b) $\displaystyle\int \cos(7x + 3)\, dx$

Problem 9.

(a) $\int x^2 \cos(x^3)\, dx$

(b) $\int x\sqrt{2x+1}\, dx$

Problem 10.

(a) $\int \sin^2 x\, dx$

(b) $\int (1 - 2\sin^2 x)\sin(2x)\, dx$

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Answer to selected problems

1. -2
2. (a) 0, (b) -8
5. (a) $x^3 + 1$, (b) $-3x\sin(x)$
6. (a) $2x\cos(x^2)$, (b) $-\dfrac{2x}{1+x^2}$
9. (a) $\dfrac{1}{3}\sin(x^3) + C$, (b) $\dfrac{1}{10}(2x+1)^{5/23} - \dfrac{1}{6}(2x+1)^{3/2} + C$